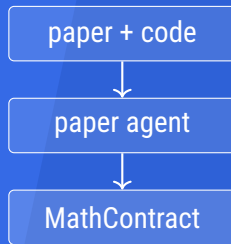


From Papers to Math Contracts

AgtXlv / MathContractRegistry: auditable claim reuse

Math-first pilot

AgtXlv / MathContractRegistry / 2026



Start with a concrete promise

For a theorem query, return more than a list of “related papers”:
an auditable, reusable dependency path with gaps marked.

target address

where the target claim lives in the
registry

dependency path

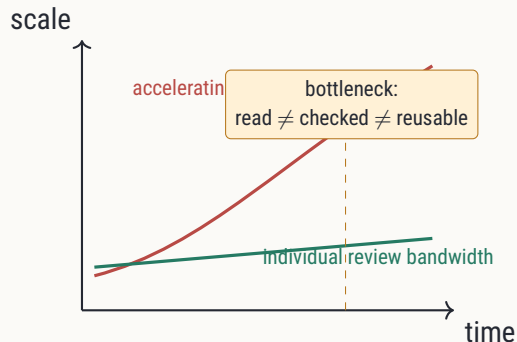
which definitions, assumptions, and
theorems it actually uses

frontier + delta

connected evidence, unresolved
boundary, and new work needed

Core object: a reasoning-centric, versioned, claim-level MathContract.

Reading bandwidth becomes the bottleneck

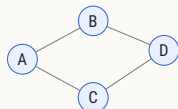


- Finding a paper is no longer the hard part.
- Harder: which results can later work safely rely on?
- A reader must track **definitions, quantifiers, domains, proof dependencies, and counterexamples.**

AgtXlv changes the unit of reading:
from a whole PDF to an inspectable
mathematical interface.

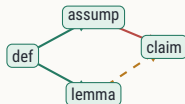
A citation graph is not a proof graph

Citation graph



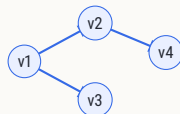
Paper *A* cites *B*,
but for what reason?

Scientific claim graph



Which premises
does the claim actually use?

Lean support graph

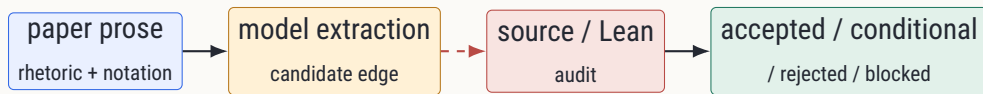


Which declarations
were checked in a fixed environment?

A cites B $\nRightarrow$ A uses a theorem from B, or that B's assumptions hold.

AgtXlv tracks provenance, mathematical dependencies, Lean support, and separate computational and physical evidence.

Automated semantic extraction creates false edges



- rhetorical overstatement or missing assumptions
- notation, quantifier, or domain mismatch
- background citation misread as a load-bearing dependency
- honest intermediate mistakes or literature disputes
- hidden counterexamples or errata
- a model can inherit only the text it reads

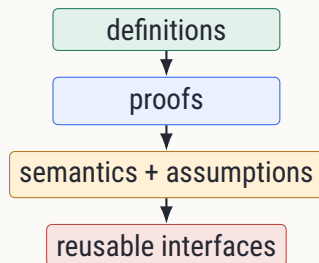
Therefore the first graph is ORACLE_PROPOSED / CANDIDATE **and cannot promote itself to accepted.**

Current edge evidence remains Oracle-proposed, unreviewed at edge level, and candidate.

Formal physics infrastructure is becoming usable

Representative direction	arXiv
HepLean	2405.08863
Physics index notation	2411.07667
PhysProver	2601.15737
MerLean	2602.16554
Lean-QEC	2605.16523
TNLean	2607.07857
Lean-QIT	2607.09632
Axioms / Seiberg–Witten	2607.06379

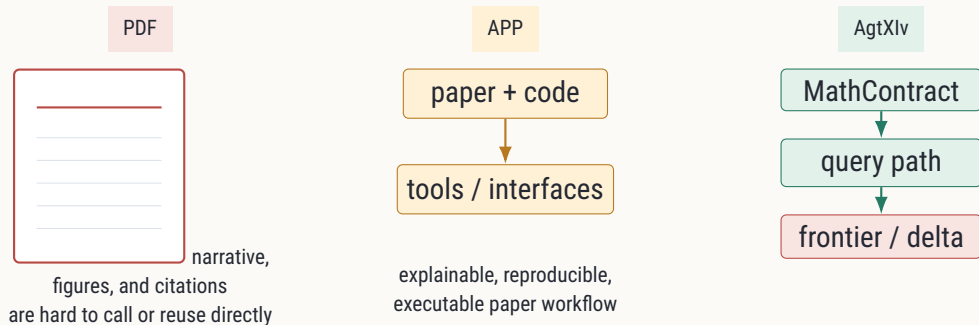
arXiv identifiers checked; kernel-checked proofs still need alignment with the intended statement.



Intuition:

formalization is not merely formula-to-code translation; it exposes missing premises, intent drift, and broken literature bridges.

A paper agent turns a static PDF into a callable object



Sirui Lu and Xiao-Liang Qi's **Agentic Publication Protocol (APP)** [arXiv:2606.27386](#) packages a versioned repository, code, data, environment, and AGENTS.md as one publication object. AgtXlv adds claim-level evidence for why a result can be inherited.

Related direction: Paper2Agent, [arXiv:2509.06917](#).

APP makes a repository a stable publication object

DEFINITION

publication object = public git repository + a specific release

Three layers

1. Human publication

manuscript / README / license / code / data / environment



2. Agent-facing layer

AGENTS.md: ground truth / navigation / setup



3. Release + verification

tag / commit-tree hash / manifest / author approval

contents / operator manual / version seal

Five-step publish-paper flow

RUN / PACK / INSTRUCT / INSPECT / SEAL

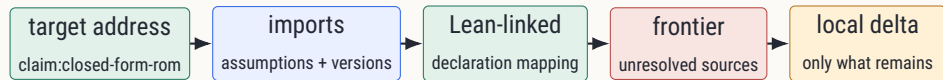
- 1 reproduce-results reproduction report
- 2 prepare-staging organized public tree
- 3 define-paper-agent concise AGENTS.md
- 4 validate-publication structural + usability checks
- 5 release-outcome approved tag + manifest

The detailed records distinguish reproduced, blocked, manual-only, inconsistent, and privacy or path failures.

Validation checks structure, metadata, and reader usability; it is not peer review or a certificate of scientific truth. **AgtXiv adds claim-level dependencies and evidence on top.**

Sirui Lu and Xiao-Liang Qi, Agentic Publication Protocol, arXiv:2606.27386.

A query should return a path receipt



Auditable fields in the current registry

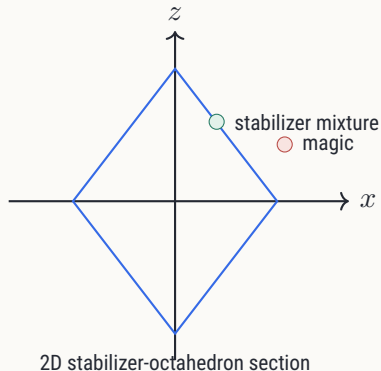
- normalized statement, quantifiers, domain, and exactness
- source anchors, imports, and verification references
- Lean declaration, blockers, and versions

Not acceptable

- returning only a list of “related papers”
- describing a candidate edge as an accepted dependency
- using fluent prose to hide a missing premise

A MathContract is a reusable mathematical interface with assumptions, sources, formal links, and evidence state.

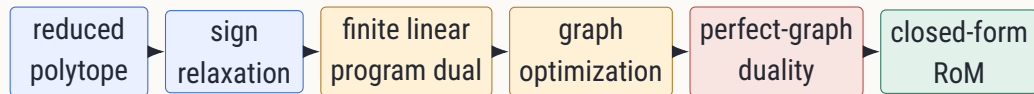
Physical picture: a measurement window sees only a shadow



- Six Pauli eigenstates form the one-qubit stabilizer octahedron.
- A small measurement window casts a shadow of the full polytope.
- Outside: a sound magic certificate.
Inside: unobserved magic may remain.
- RoM is the minimum l_1 cost of a signed decomposition, not a Euclidean distance.

G_M is a different object: vertices are Pauli operators, edges mean anticommutation, and independent sets are commuting contexts.

Concrete query: from a measurement window to a closed form



Question in plain language

Why, when there is no Pauli-active dependency and G_M is a perfect graph, does $\text{RoM}_M(\rho) = \max\left(1, \max_Q \sum_{P \in Q} |\text{tr}(P\rho)|\right)$?

inherited stabilizer form, projection semantics,
selected local Lean declarations

unresolved frontier weighted perfect-graph duality,
finite linear-program strong duality, source
alignment

Target contract: `claim:closed-form-rom`; current state: `LOCAL_DERIVATION_WITH_BLOCKED_IMPORTS`.

The registry is explicit about what remains unverified

whole-paper nodes	74
mathematical interfaces	65
mapped Paper Agents	5
typed candidate edges (demo)	114
unresolved external interfaces	8

The canonical whole-paper graph has 129 edges; the demo keeps 114 edges between mathematical interfaces.

Lean-linked declarations	10
GAP	45
PLANNED_DELTA	2
EXTERNAL_OR_UNMAPPED	8
NOT_APPLICABLE	9

Green means a declaration mapping exists; it does not imply statement alignment or scientific acceptance.

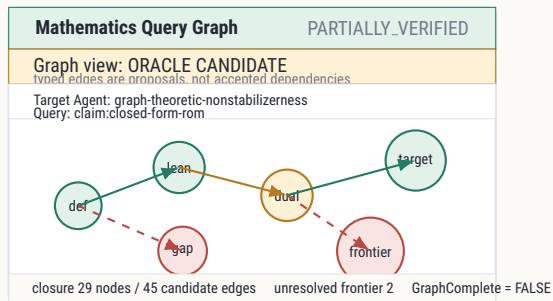
`graph_view = ORACLE_CANDIDATE`

all displayed edges remain candidate

`acceptance_effect = NONE_FROM_REGISTRY`

Counts: manifest.json, contracts.json, and demo/data/claim-dag.en.json.

Each query iteration leaves an evidence trail



Legend

- green: linked to a Lean declaration
- blue: definition / theorem interface
- yellow: candidate / scope / local delta
- red: gap, frontier, or known counterexample

Red is not one failure mode:
distinguish GAP, PROOF_GAP, BLOCKED,
and FALSE.

Iteration 0: split a compound theorem into interfaces

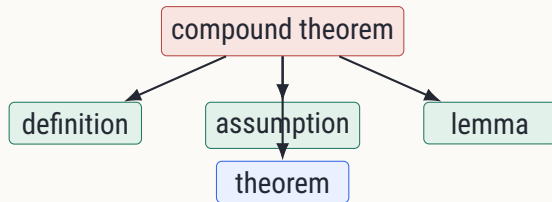
Input

One compound theorem mixes a projected polytope, sign constraints, linear-program duality, graph equivalence, and a closed form in one paragraph.

Every node carries at least:

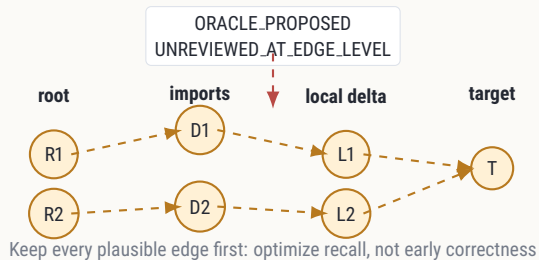
- source anchor
- normalized statement
- quantifiers / domain / assumptions
- exactness and scope
- downstream blockers

Keep undefined claims explicit; do not fill gaps with prose.



AgtXlv Mathematics Pipeline: strict mathematical claim decomposition.

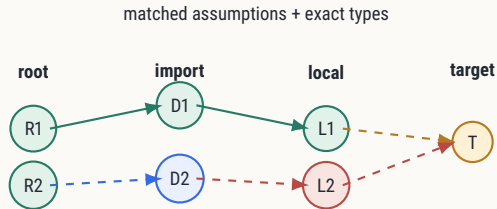
Iteration 1: build a high-recall candidate graph



green / red are first-pass display states, not scientific acceptance.

graph.view = ORACLE.CANDIDATE; edges remain candidates until source and Lean alignment.

Iteration 2: align sources and connect Lean declarations



- green: linked to a Lean declaration
- blue: source-aligned, but no reusable Lean bridge yet
- red: GAP / unresolved / blocked
- yellow: target's local delta is still unimplemented

This answers what connects formally, not whether the physics and source meanings fully agree.

Lean-linked means declaration mapping, not verified coverage.

Iteration 3: critical audit separates failure modes

V-representation: proof gap

PROOF_GAP

THEOREM_NOT_REFUTED

CONDITIONAL_REPAIR

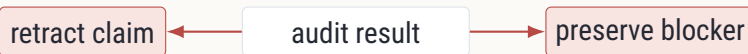
An intermediate bijection in the appendix fails. The registry records an independent repair route and two open premises; it does not claim the theorem is refuted.

Fixed window: counterexample

FALSE

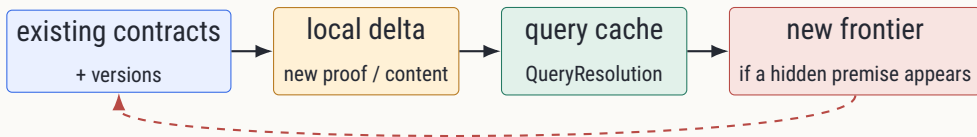
EXPLICIT_COUNTEREXAMPLE

For a fixed measurement window M , a one-qubit Clifford counterexample increases reduced RoM. That is different from a V-representation proof gap.



Audit facts: Stabilizerness/AGTXIV_IMPLEMENTATION.md and registry blocker/status fields.

Iteration 4: prove only the local delta, then recompute closure



Engineering intuition

- A new paper is an increment to a knowledge base, not an isolated PDF.
- Verified roots need not be rebuilt.
- Implement only the new proof or content needed for the current target.

Non-monotonicity matters

- Closing a proof can expose a hidden premise.
- The frontier need not shrink every round.
- A smaller graph is not the same as truer science.

The end state is not one Boolean

Dependency graph complete?

FALSE

An unresolved external mathematical frontier remains.

Verification closed?

FALSE

Source, Lean, computational, and physical evidence still need separate acceptance.

The registry can return `PARTIALLY_VERIFIED` instead of inventing a global `VERIFIED` state.

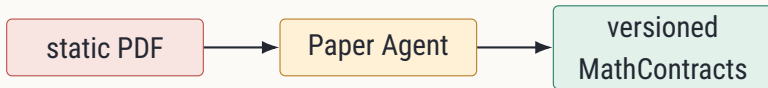
Four independent evidence axes

1. source fidelity: what did the source claim?
2. mathematical correctness: do the assumptions imply the result?
3. computational reproducibility: do code, data, and environment reproduce it?
4. physical-semantic alignment: does the formal object match the intended physics?

Lean kernel: the conclusion follows under the encoded definitions and assumptions.

Takeaway

A new paper is an increment to a large knowledge base.



Prefer an incomplete path with a visible red frontier
to a fluent, unauditable “verified” answer.

AgtXlv: verification-aware incremental search over scientific claims.

Appendix: arXiv source packages used in this demo

- Agentic Publication Protocol: `arXiv:2606.27386`
Source unpacked under `Reference/Agentic Publication Protocol An Attempt to Modernize Scientific Publication/`.
- Multi-agent Autoformalization of Tensor Network Theory: `arXiv:2607.07857`
Source unpacked under `Reference/Multi-agent Autoformalization of Tensor Network Theory/`.
- Lean-QIT: `arXiv:2607.09632`
Source unpacked under `Reference/Lean-QIT Towards a Formal Infrastructure for Quantum Information Theory/`.
- End-to-End Formalization of Quantum Error Correction: `arXiv:2605.16523`
Source unpacked under `Reference/End-to-End Formalization of Quantum Error Correction/`.

Related formal-physics identifiers: 2405.08863, 2411.07667, 2601.15737, 2602.16554, 2603.08139, 2607.06379.

The list avoids presenting “formal infrastructure” as one library or one verification standard.